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Genome wide identification of *Dof* transcription factor gene family in sorghum and its comparative phylogenetic analysis with rice and Arabidopsis

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Abstract The Dof (DNA binding with One Finger) family represents a classic zinc-finger transcription factors involved with multifarious roles exclusively in plants. There exists great diversity in terms of number of *Dof* genes observed in different crops. In current study, a total of 28 putative *Dof* genes have been predicted in silico from the recently available whole genome shotgun sequence of *Sorghum bicolor* (L.) Moench (with assigned accession numbers TPA:BK006983–BK007006 and TPA:BK007079–BK007082). The predicted *SbDof* genes are distributed on nine out of ten chromosomes of sorghum and most of these genes lack introns based on canonical intron/exon structure. Phylogenetic analysis of 28 SbDof proteins resulted in four subgroups constituting six clusters. The comparative phylogenetic analysis of these Dof proteins along with 30 rice and 36 Arabidopsis Dof proteins revealed six major groups similar to what has been observed earlier for rice and Arabidopsis. Motif analysis revealed the presence of conserved 50–52 amino acids Dof domain uniformly distributed across all the 28 Dof proteins of sorghum. The in silico *cis*-regulatory elements analysis of these *SbDof* genes suggested its diverse functions associated with light responsiveness, endosperm specific gene expression, hormone responsiveness, meristem specific expression and stress responsiveness.

Keywords Dof · Sorghum · Rice · Arabidopsis · Alignment · Phylogenetic tree · Motif · *Cis*-regulatory element

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Introduction

In general many biological processes are strictly regulated through transcriptional control of particular genes in plants. Therefore, plants need a large number of transcription factors (TF) for proper and strict transcriptional regulation in response to developmental stages and environmental changes. Some classes of plant transcription factors have DNA-binding domains similar to animal transcription factors, whereas other classes of transcription factors appear to have specifically evolved in plants [1]. A family of TFs putatively specific to plants is the Dof family which has been extensively reviewed [1–7]. Dof proteins are mostly 200–400 amino acids long with a conserved DNA binding Dof domain of 50–52 amino acid residues which is structured as a Cys2/Cys2 zinc finger recognizing a *cis*-regulatory element with the common core sequence 5′-AAAG-3′ [5, 8, 9]. The Dof domain is a bifunctional domain that mediates both DNA–protein and protein–protein interactions [10, 11].

Dof proteins contain a conserved Dof domain located at the N-terminal region and a transcriptional regulation domain at the C-terminus. Serine stretches that are frequently located immediately downstream of the Dof domains might be the molecular hinges linking two domains [5]. Dof transcription factors have been reported to be associated with regulation of vital processes in plants such as photosynthetic carbon assimilation, light regulated gene expression, accumulation of seed storage proteins, germination, dormancy, response to phytohormones, flowering time, and guard cell-specific gene expression [6].

There exists a great diversity in terms of number of *Dof* genes in plants. Various in silico analysis have predicted 30, 36, 24, 31 and 54 *Dof* genes in rice, Arabidopsis, barley, wheat and maize respectively [6, 12–14]. The phylogenetic relationships between rice and Arabidopsis Dof proteins revealed the presence of four major clusters of orthologous genes (MCOGs) [6]. Similar types of phylogenetic analyses have been reported for other plant TF families such as the WRKY, R2R3–MYB, bZIP, MADS, GATA, etc. [15–19].

Based on phylogenetic analysis of Dof sequences across the representative organisms belonging to green unicellular algae to vascular plants, the origin and evolution of the Dof transcription factor family has been reported. A total of 116 *Dof* genes representing green unicellular alga (*Chlamydomonas reinhardtii*), moss (*Physcomitrella patens*), fern (*Selaginella moellendorffii*), gymnosperm (*Pinus taeda*), dicotyledonous angiosperm (*Arabidopsis thaliana*), and monocot (*Oryza sativa* and *Hordeum vulgare*) have been characterized in silico. The phylogenetic tree analysis revealed the existence of six major clusters of orthologous and paralogous genes that probably originated by gene duplication events from a paraphyletic basal grade [12].

There have been substantial studies on *Dof* genes of Arabidopsis and rice representing the model crop for dicot and monocot groups. In Arabidopsis, some of the well characterized *Dof* genes includes *DAG1* and *DAG2* genes which are associated with seed germination [20–22], *CDF1*, *CDF2* and *CDF3* genes which are involved with photoperiodic control of flowering [23, 24], *COG1* and *OBP3* regulate phytochrome signaling [25, 26] and *OBP1* is involved in the cell cycle regulation [27]. In rice, *OsDof3* is involved in the gibberellin-regulated expression [28, 29]. Maize *Dof1* has been shown to be an activator for gene expression associated with carbohydrate metabolism, including the phosphoenolpyruvate carboxylase gene [30, 31] while maize *Dof2* appears to be a repressor that can block transactivation of *Dof1* [32]. In wheat, *TaDof1* is associated with carbon metabolism [33].

A prolamin-box binding factor (PBF) Dof associated with regulation of endosperm-specific seed storage proteins have been studied in maize, barley, wheat and finger millet [34–39]. There are some Dof proteins which are involved in responses to plant hormones [40–43], in regulation of defense gene expression [44–46] and in stomata guard cell function [47].

Cereals including rice, maize, wheat, barley, rye, sorghum, oats and millets are considered to be the most important group of cultivated plants in terms of food production and acreage covered, providing most of the calories and proteins requirement of our daily diet [48]. These cereals comprise more than 10,000 species belonging to family Poaceae with several subfamilies like Oryzoideae, Pooideae, Panicoideae and Chloridoideae [49]. Though there exists great diversity among cereals in terms of genome size, ploidy level and chromosome numbers, attempts have been made to reveal the existing synteny and colinearity on the basis of comparative genomics [50–54].

Sorghum is the fifth most important cereal crop in the world based on total grain production. Taxonomically, *Sorghum bicolor* (L.) Moench subs. *bicolor* has 15 races: the 5 basic races of bicolor, durra, caudatum, kafir, and guinea, and their 10 intermediates. Sorghum has unique properties that make it well suited for food uses [55].

Sorghum has a relatively small genome (750 Mbp), extraordinarily diversity of germplasm and incremental divergence from maize and rice, which greatly aid the discovery and analysis of grass genes through comparative genomics. Besides, sorghum is a C4 plant while rice, the first fully sequenced cereal genome is more representative of C3 photosynthetic grasses. Drought tolerance, stalk reserve retention, potential to produce more grain per unit area makes sorghum an attractive system for the study [56].

This paper reports genome wide in silico identification of putative *SbDof* gene family of *S. bicolor* (L.) Moench from the recently available whole genome shotgun sequence of sorghum, its annotation for chromosomal location(s), gene structure and phylogenetic tree construction. Further its comparative phylogeny with rice and Arabidopsis *Dof* gene family has been attempted. The putative functions of the predicted *SbDof* genes were investigated by analyzing the *cis*-regulatory elements associated with these genes in the promoter region.

Materials and methods

Database searches for the identification of Dof family members in *S. bicolor* (L.) Moench

The Dof sequences of *O. sativa* and *A. thaliana* were collected from two different sources, DRTF (http://drtf.cbi.pku.edu.cn/) [57] and DATF (http://datf.cbi.pku.cn) database respectively [58]. The nucleotide sequences of Dof domain were used to search the potential Dof domain homologs hit in the whole genome shotgun sequence of *S. bicolor* through BLASTN, TBLASTN and discontinuous MEGABLAST [59, 60] at the NCBI database (http://www.ncbi.nlm.nih.gov) [61]. The 3 kb upstream and 3 kb downstream sequences of Dof domain homologs were retrieved from whole genome shotgun sequence of *S. bicolor* for fishing out the putative *SbDof* genes. The annotated sequences were further subjected to bioinformatics servers namely FGENESH [62] and GENESCAN [63] for prediction of full length genes with putative CDS and protein sequences. The putative Dof protein sequences of *S. bicolor* were subjected to protein functional analysis using PFAM version 24.0 [64], PROSITE version 20.58 [65], INTERPROSCAN version 24 [66], and MOTIFSCAN [67] databases. The Dof protein sequences were then subjected to NUCPRED server [68] for the identification of nuclear localization signal (NLS).

Mapping of *SbDof* genes on sorghum chromosomes and its intron/exon gene structure prediction

Each of the *SbDof* genes was positioned on sorghum chromosomes by the BLASTN search with NCBI genomes (chromosome) database. The resulting position of *SbDof* genes on sorghum chromosome accession numbers CM000760–CM000769 were manually marked on bar. The gene structure of predicted *SbDof* genes were carried out using FGENESH server and intron/exon structures were manually designed.

Dof protein alignment and phylogenetic analysis

The identified SbDof proteins were aligned using ClustalX2.0.10 [69]. The phylogenetic tree was inferred by bootstrap (1,000 reiterations), neighbor joining [70] phylogenetic inference using ClustalX2.0.10.

Identification of conserved motif and *cis*-regulatory element analysis

The deduced protein sequences of the 28 *SbDof* genes of sorghum along with 36 AtDof proteins of Arabidopsis and 30 OsDof proteins of rice were analyzed by means of the MEME (Multiple EM for Motif Elicitation) program software version 4.4.0 [71, 72] for motif analysis. To identify conserved motifs in these sequences, the selection of maximum number of motif was set to 50 with minimum and maximum width of 20 and 50 amino acids respectively while other factors were of default selections. For promoter analysis 500 bp upstream sequences from the initiation codon of the putative *SbDof* genes were retrieved. The retrieved sequences were then subjected to search for CARE program (http://bioinformatics.psb.ugent.be/webtools/plantcare/html/) of PlantCARE database [73] for identification of *cis*-regulatory elements.

Results and discussion

Genome wide identification of *Dof* gene family

Recently sequenced genome of *S. bicolor* [56] provided an opportunity to predict the complete set of non redundant *Dof* genes using various bioinformatics tools. The nucleotide and amino acid sequences of the conserved Dof domain were used to perform BLAST search and a total of 28 *Dof* transcription factor genes were annotated from the whole genome of sorghum. The predicted sequences of 28 *SbDof* genes were then submitted to Third Party Annotation (TPA) Section of the DDBJ/EMBL/GenBank databases and were assigned the accession numbers TPA:BK006983–BK007006 and TPA:BK007079–BK007082 (Table 1).

The presence of the conserved Dof domain in the predicted SbDof protein was a typical feature for consideration as member of *Dof* family of transcription factor. Protein functional analysis of these 28 putative SbDof proteins using INTERPROSCAN has shown identity with InterPro accession number IPR003851. These proteins also showed similarity with signature accession namely PD007478 (ProDom database), PF02701 (Pfam database), PS01361 (PROSITE patterns database) and PS50884 (PROSITE profiles database) confirming their identity to Dof like proteins. The deduced protein sequences of *SbDof* genes were then subjected to motif scan database. Motif scan integrated with PeroxiBase profiles, PROSITE patterns, PROSITE profiles, HAMAP profiles, Pfam HMMs (local models), Pfam HMMs (global models) and PeroxiBase profile databases revealed the presence of glycine, threonine, alanine, serine, proline, histidine, glutamine, methionine and asparagine rich regions in different *SbDof* genes. A typical zinc finger Dof type profile was observed in all *SbDof* genes.

The SbDof4 subjected to INTERPROSCAN for protein functional analysis revealed its identity with InterPro accession number IPR001412 aminoacyl-tRNA synthetase class I conserved site and PS00178 (PROSITE profiles database) indicating an additional function of t-RNA aminoacylation associated with protein translation. Only three SbDof proteins namely SbDof1, SbDof7 and SbDof21 showed nuclear localization signal (NLS) when subjected to NUCPRED software. The NLS signal for SbDof1 was found to be RKRPRPR while SbDof7 and SbDof21 showed KRRRV as NLS signal.

Chromosomal locations of predicted *SbDof* genes

The predicted 28 *Dof* genes are distributed on nine chromosomes as provided in the chromosome data of *S. bicolor* in GenBank [56]. Chromosome 1 and 3 have a maximum of 7 and 6 *Dof* genes respectively, while chromosomes

Table 1 In silico predicted *Dof* genes from whole genome shotgun sequence of *S. bicolor* (L.) Moench

Annotated gene	Source	Assigned accession number	Chromosome number	Gene structure
SbDof1	ABXC01005623	TPA:BK007079	8	Intronless
SbDof2	ABXC01000338	TPA:BK007080	1	One intron
SbDof3	ABXC01000645	TPA:BK006983	1	Intronless
SbDof4	ABXC01000029	TPA:BK006984	1	Intronless
SbDof5	ABXC01000170	TPA:BK006985	1	Intronless
SbDof6	ABXC01001528	TPA:BK006986	2	Intronless
SbDof7	ABXC01001444	TPA:BK006987	2	Intronless
SbDof8	ABXC01001648	TPA:BK006988	2	Intronless
SbDof9	ABXC01000156	TPA:BK006989	1	Intronless
SbDof10	ABXC01000069	TPA:BK006990	1	Intronless
SbDof11	ABXC01002298	TPA:BK006991	3	Intronless
SbDof12	ABXC01002251	TPA:BK006992	3	Intronless
SbDof13	ABXC01002199	TPA:BK006993	3	Intronless
SbDof14	ABXC01001839	TPA:BK006994	3	One intron
SbDof15	ABXC01001828	TPA:BK006995	3	Two introns
SbDof16	ABXC01002936	TPA:BK006996	4	Intronless
SbDof17	ABXC01002910	TPA:BK006997	4	One intron
SbDof18	ABXC01002897	TPA:BK006998	4	Intronless
SbDof19	ABXC01003020	TPA:BK006999	5	Intronless
SbDof20	ABXC01004404	TPA:BK007000	6	Intronless
SbDof21	ABXC01004473	TPA:BK007001	6	One intron
SbDof22	ABXC01005057	TPA:BK007002	7	Intronless
SbDof23	ABXC01000714	TPA:BK007003	1	One intron
SbDof24	ABXC01005079	TPA:BK007004	8	One intron
SbDof25	ABXC01006212	TPA:BK007005	9	Intronless
SbDof26	ABXC01005734	TPA:BK007006	9	Intronless
SbDof27	ABXC01001715	TPA:BK007081	3	Intronless
SbDof28	ABXC01005582	TPA:BK007082	8	Intronless

number 2, 4 and 8 have 3 *Dof* genes each (Table 1). Chromosome 10 does not contain any *Dof* gene. The distribution of *Dof* genes among the 10 chromosomes of sorghum is shown in Fig. 1.

The distribution of *Dof* genes in rice and Arabidopsis chromosomes has been studied [6]. In case of rice, 30 *Dof* genes were found to be organized on 11 out of the 12 chromosomes, with maximum number of 6 *Dof* gene distributed on both chromosomes 1 and 3 (Table 2) while 36 *Dof* genes of Arabidopsis were distributed among all the five chromosomes with maximum nine *Dof* gene on chromosome 1 (Table 3).

Phylogenetic relationships among Dof proteins of sorghum

The complete catalog of Dof protein in a single plant species is useful for viewing the existing structural and functional diversity associated with its diverse roles played in plants. The evolutionary relationship between different Dof proteins were analyzed by subjecting the deduced amino acid sequences of the identified 28 *SbDof* genes for multiple sequence alignment. A well conserved four cysteine residues putatively involved in the formation of the zinc finger structure (Fig. 2) was observed when Dof domain sequence of different SbDof proteins were aligned.

The Dof domain of sorghum revealed highly conserved sequences with 25 out of 52 amino acids being 100% conserved in all the 28 SbDof proteins while 9 amino acids showed variation in only two amino acid residues. The phylogenetic tree constructed using ClustalX2.0.10 software with bootstrapping revealed the existence of two major groups A and B. Group A was further divided in two subgroups A1 and A2. Group B was similarly divided into two subgroups B1 and B2 which were further arranged into two branches as shown in Fig. 3a. As a result the Dof family in sorghum exists in four subgroups comprising of six clusters. A very high bootstrap value suggested common origin for *Dof* genes of each sub-group.

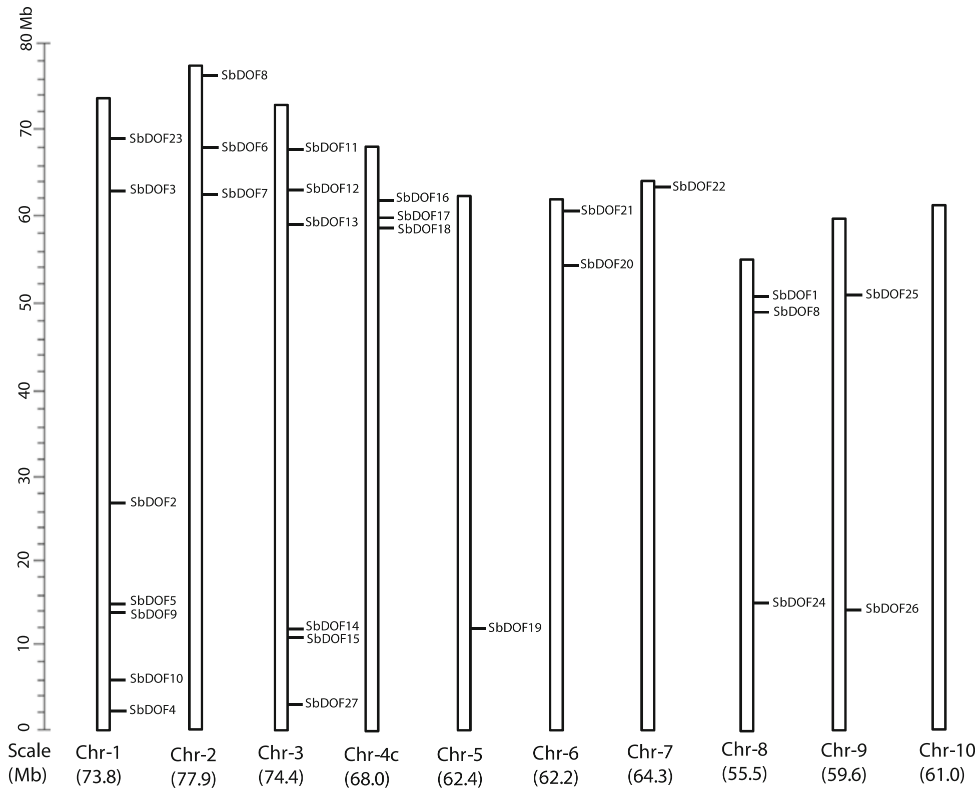

Fig. 1 Chromosomal localization of 28 *SbDof* genes on ten chromosomes of sorghum. The chromosome numbers and its size are indicated at the *bottom* of each bar. Size of chromosome represented using vertical scale

Gene structure prediction

The putative gene structure of the predicted *SbDof* gene family in terms of intron/exon distribution pattern is shown in Fig. 3b. A total of 21 out of 28 *SbDof* genes were found to be intronless. The *SbDof15* seems to have complex gene structure with presence of two introns, while *Dof* genes with at least one intron are also observed in sorghum. Most members of *SbDof* genes with more than one exon represent sub groups A1 in the phylogenetic tree (Fig. 3a, b). A total of 17 out of 36 *Dof* gene models in rice (Table 2) and 21 out of 43 *Dof* gene models in Arabidopsis are intronless (Table 3). There exists great similarity among *Dof* genes of rice, Arabidopsis and sorghum in terms of gene structure with a majority of *Dof* genes either lacking intron or having single intron [6].

Evolutionary relationships of sorghum, rice and Arabidopsis with respect to *Dof* gene family

The predicted 28 SbDof proteins were subjected to multiple sequence alignment along with 36 Arabidopsis and 30 rice Dof proteins and a phylogenetic tree was constructed using software ClustalX2.0.10 with neighbor joining method and bootstrap analysis (1,000 reiterations) (Fig. 4). A total of six major groups of Dof proteins designated as A, B, C, D, E and F were observed for sorghum similar to what has been reported earlier for Dof proteins of wheat, rice and Arabidopsis [13].

The SbDof8, SbDof14 and SbDof15 of sorghum shows maximum similarity with AtDof17, AtDof30 and AtDof33 of Arabidopsis representing major group-A, which are basically CDF proteins associated with regulation of photoperiodic control of flowering [23, 74]. The Arabidopsis Dof proteins namely AtDof26, AtDof24, AtDof25 and AtDof23 constitute a distinct major group-B similar to what has been reported earlier in Arabidopsis cluster C3 in MCOG Cc [6]. These sets of *Dof* genes might be exclusively present in Arabidopsis as no apparent counterpart has been observed in rice or sorghum. In the major group-C, the SbDof22, SbDof7 and SbDof1 show similarity with rice OsDof18, OsDof30 and OsDof27 which are basically MNb1a protein. In group-D, SbDof21 shows maximum similarity with AtDof18 which is an Arabidopsis OBP1 protein involved in defense response [44]. The other members of group-D i.e. SbDof24 and SbDof19 show similarity with rice OsDof23 (RPBF) which is associated with the regulation of seed storage proteins [28, 29, 38]. In group-F the SbDof16 and SbDof20 form cluster with AtDof14 and AtDof21, an Arabidopsis DAG proteins (Dof affecting germination proteins) indicating that these proteins might be involved in seed germination [20, 22]. The overall phylogenetic analysis of sorghum Dof proteins clearly indicates its maximum similarity with rice Dof

Table 2 List of *Dof* genes of *O. sativa* downloaded from http://drtf.cbi.pku.edu.cn/

Gene	Locus	Gene model	Chromosome number	Gene structure
OsDof1	Os07g48570	Os07g48570.1	7	One intron
OsDof2	Os01g55340	Os01g55340.1	1	Intronless
OsDof3	Os01g15900	Os01g15900.1	1	One intron
OsDof4	Os01g17000	Os01g17000.1	1	One intron
OsDof5	Os03g16850	Os03g16850.1	3	One intron
OsDof6	Os03g07360	Os03g07360.1	3	One intron
OsDof7	Os10g26620	Os10g26620.1	10	One intron
OsDof8	Os05g36900	Os05g36900.1	5	Intronless
OsDof9	Os01g64590	Os01g64590.1	1	Intronless
OsDof10	Os07g32510	Os07g32510.1	7	Intronless
OsDof11	Os02g47810	Os02g47810.1	2	One intron
OsDof12	Os02g49440	Os02g49440.1	2	Intronless
OsDof13	Os03g38870	Os03g38870.1	3	Intronless
OsDof14	Os06g17410	Os06g17410.1	6	Intronless
OsDof15	Os01g09720	Os01g09720.1	1	Intronless
OsDof16	Os04g47990	Os04g47990.1	4	One intron
		Os04g47990.2		Intronless
OsDof17	Os02g45200	Os02g45200.1	2	One intron
		Os02g45200.2		Two introns
		Os02g45200.3		Intronless
		Os02g45200.4		One intron
OsDof18 (*MNB1A*)	Os08g38220	Os08g38220.1	8	Intronless
OsDof19	Os03g42200	Os03g42200.1	3	Two introns
OsDof20	Os01g48290	Os01g48290.1	1	Intronless
OsDof21	Os12g38200	Os12g38200.1	12	One intron
OsDof22	Os07g13260	Os07g13260.1	7	Intronless
OsDof23 (*RPBF*)	Os02g15350	Os02g15350.1	2	One intron
OsDof24	Os05g02150	Os05g02150.1	5	One intron
OsDof25	Os04g58190	Os04g58190.1	4	One intron
		Os04g58190.2		Two introns
		Os04g58190.3		Two introns
OsDof26	Os10g35300	Os10g35300.1	10	One intron
OsDof27 (*MNB1A*)	Os12g39990	Os12g39990.1	12	Intronless
OsDof28	Os03g55610	Os03g55610.1	3	Intronless
OsDof29	Os03g60630	Os03g60630.1	3	Intronless
OsDof30 (*MNB1A*)	Os09g29960	Os09g29960.1	9	Intronless

proteins rather than Arabidopsis owing to its monocot nature. The evolutionary relationship among the rice *Dof* gene family has been attempted based on their DNA binding domain sequences which revealed four major groups two of which were further divided in two subgroups resulting into a total of seven subgroups [6]. Similarly phylogenetic analysis of a total of 116 genes in different organisms from green unicellular algae to vascular plant including 30 rice *Dof* genes also revealed seven subfamily [12]. Thus the subgroups of rice *Dof* genes are more than sorghum which might be due to larger number of *Dof* genes in rice.

Motif analysis

The 28 Dof proteins of sorghum along with the 66 Dof proteins of Arabidopsis and rice were analyzed for the presence of conserved motifs by means of MEME software. A total of 50 conserved motifs were observed in all the 94 Dof proteins (Fig. 5). The motif 1 uniformly observed in all the Dof proteins except AtDof33 of Arabidopsis represents the conserved Dof domain of 50 amino acids. The lack of motif 1 in AtDof33 might be due to absence of some sequences at the N-terminus of Dof domain. There were number of common motifs

Table 3 List of *Dof* genes of *A. thaliana* downloaded from http://datf.cbi.pku.cn

Gene name	Locus	Gene model	Chromosome number	Gene structure
AtDof1 (*OBP2*)	AT1G07640	AT1G07640.1	1	Intronless
		AT1G07640.2		One intron
		AT1G07640.3		One intron
AtDof2	AT1G21340	AT1G21340.1	1	Intronless
AtDof3	AT1G26790	AT1G26790.1	1	Two introns
AtDof4	AT1G28310	AT1G28310.1	1	Intronless
		AT1G28310.2		One intron
AtDof5 (*COG1*)	AT1G29160	AT1G29160.1	1	Intronless
AtDof6	AT1G47655	AT1G47655.1	1	Intronless
AtDof7 (*ADOF1*)	AT1G51700	AT1G51700.1	1	Intronless
AtDof8	AT1G64620	AT1G64620.1	1	One intron
AtDof9	AT1G69570	AT1G69570.1	1	One intron
AtDof10	AT2G28510	AT2G28510.1	2	One intron
AtDof11	AT2G28810	AT2G28810.1	2	One intron
AtDof12	AT2G34140	AT2G34140.1	2	Intronless
AtDof13	AT2G37590	AT2G37590.1	2	One intron
AtDof14 (*DAG2*)	AT2G46590	AT2G46590.1	2	Intronless
		AT2G46590.2		One intron
AtDof15 (*ADOF2*)	AT3G21270	AT3G21270.1	3	Intronless
AtDof16	AT3G45610	AT3G45610.1	3	One intron
AtDof17 (*CDF3*)	AT3G47500	AT3G47500.1	3	One intron
AtDof18 (*OBP1*)	AT3G50410	AT3G50410.1	3	Intronless
AtDof19	AT3G52440	AT3G52440.1	3	Intronless
AtDof20 (*OBP3*)	AT3G55370	AT3G55370.1	3	Two introns
		AT3G55370.2		One intron
AtDof21 (*DAG1*)	AT3G61850	AT3G61850.1	3	Two introns
		AT3G61850.2		One intron
AtDof22	AT4G00940	AT4G00940.1	4	Intronless
AtDof23	AT4G21030	AT4G21030.1	4	Intronless
AtDof24	AT4G21040	AT4G21040.1	4	Intronless
AtDof25	AT4G21050	AT4G21050.1	4	Intronless
AtDof26	AT4G21080	AT4G21080.1	4	Intronless
AtDof27	AT4G24060	AT4G24060.1	4	One intron
AtDof28	AT4G38000	AT4G38000.1	4	Intronless
AtDof29	AT5G02460	AT5G02460.1	5	One intron
AtDof30 (*CDF2*)	AT5G39660	AT5G39660.1	5	One intron
		AT5G39660.2		One intron
AtDof31	AT5G60200	AT5G60200.1	5	One intron
AtDof32 (*OBP4*)	AT5G60850	AT5G60850.1	5	Intronless
AtDof33 (*CDF1*)	AT5G62430	AT5G62430.1	5	Intronless
AtDof34	AT5G62940	AT5G62940.1	5	One intron
AtDof35	AT5G65590	AT5G65590.1	5	Intronless
AtDof36	AT5G66940	AT5G66940.1	5	Intronless

observed for sorghum, rice and Arabidopsis. A total of 12 motifs designated as motif 15, 23, 25, 26, 27, 30, 33, 35, 39, 40, 43 and 47 were common for sorghum and rice, while motif 31 was observed common for rice and Arabidopsis. Motifs exclusively present in Arabidopsis are motif 14, 20, 22, 34, 38, 45, 46 and 50. In sorghum motif 24, 28 and 48 are unique as they were not observed in rice or Arabidopsis. The presence of unique motifs exclusively in a particular crop might be associated with specific function.

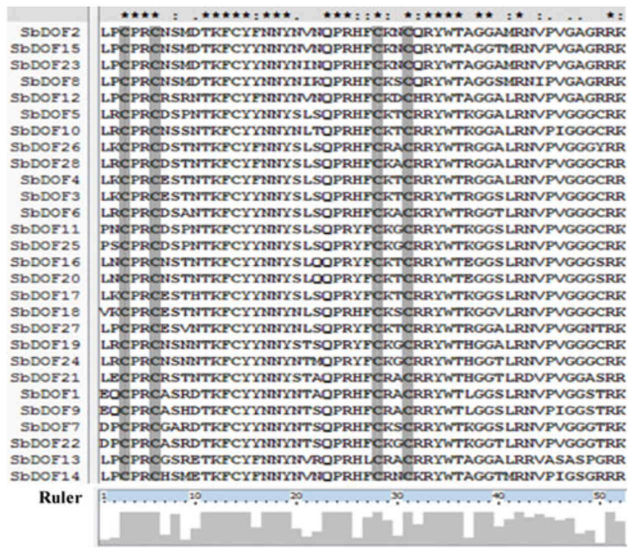

Fig. 2 Multiple sequence alignment of the Dof DNA-binding domain from the predicted *SbDof* genes containing a full-length domain. The four cysteine residues associated with zinc finger structure of *Dof* family is *shaded*. Strongly conserved residues are indicated above the alignment with *asterisks* indicating a perfectly conserved residue, with *colons* indicating the residue variation occurs with strongly conserved groups, which commonly occur, and *dots* indicating that residue variation occur within weaker conserved residue groups. The levels of amino acid conservation at each position among the SbDof members are indicated in the histogram with the highest bars representing 100% amino acid identity

Based on the groups indicated from the phylogenetic tree of Dof proteins and motifs through MEME analysis, a schematic distribution of conserved motifs among the defined gene groups is shown in Fig. 5. Each sub group has similar number and arrangement of motifs. Some motifs were located in the proteins of specific subgroups, for instance motifs 2, 3, 4, 13, 19, 26, 38, 39, 41 and 47 for group-A, motifs 14, 22 and 34 for group-B, motif 27 and 40 for group-C, motif 24 and 28 for group-D, motifs 8 and 31 for group-E and motif 17, 35, 15, 50, 16, 11 and 23 for group-F. These similarities in motif patterns might be related to similar function of each subgroup. The multilevel consensus sequence for the identified motifs is provided in Table 4.

Cis-regulatory elements analysis

The diverse functions attributed to *Dof* genes are due to interaction of these transcription factors with different conserved sequences in the promoter regions of the respective genes. The *cis*-regulatory element analysis was carried out by retrieving 500 bp upstream sequences from the initiation codon of only 26 putative *SbDof* genes. *SbDof1* gene could not be analyzed for *cis*-regulatory elements as the 500 bp upstream sequences could not be retrieved due to the unavailability of sequences after 124 bases upstream from initiation codon. The *SbDof16* was also not studied in detail due to lack of important *cis*-regulatory elements. A large number of *cis*-regulatory elements were found when the 26 *SbDof* genes were subjected to PlantCARE database. The *cis*-regulatory elements related with five important physiological phenomena i.e. light responsiveness, endosperm specific gene expression, hormone responsiveness, meristem specific expression and stress responsiveness were found to be distributed among these *Dof* genes as shown in Table 5.

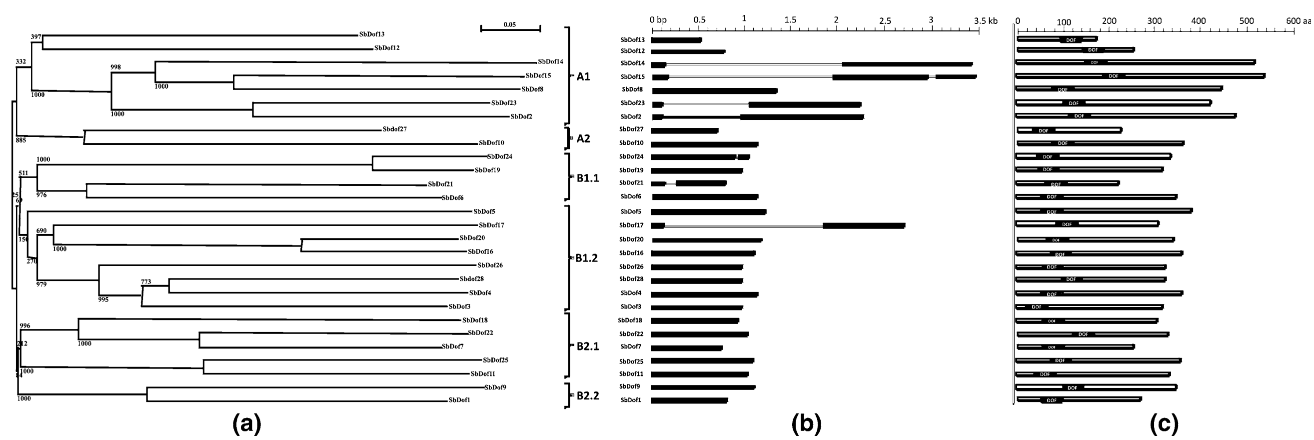

Fig. 3 **a** The phylogenetic tree of sorghum Dof proteins constructed from a complete alignment of 28 SbDof proteins by neighbor-joining methods with bootstrapping analysis. The groups and subgroups of homologous genes identified are indicated. The *scale bar* corresponds to 0.05 estimated amino acid substitutions per site. **b** Intron/exon structure of *SbDof* genes, *black bars* represents the exon while the *lines* represent the intron. The size of exons and introns can be estimated using the horizontal scale. **c** Protein structure: The Dof domain position for each SbDof protein is marked with *black box*. The length of the SbDof proteins can be estimated using the scale at *top*. aa indicate amino acid

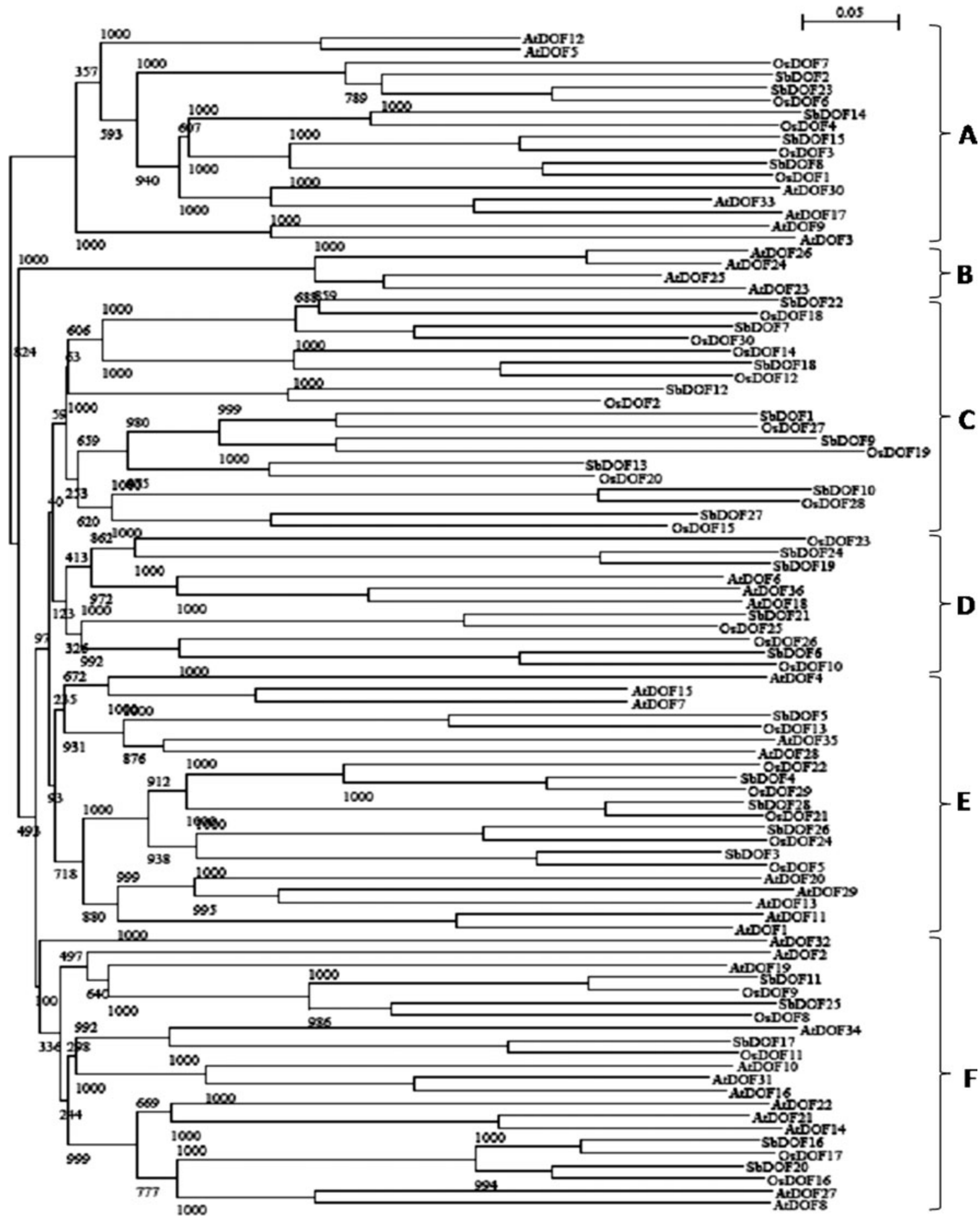

Fig. 4 Phylogenetic relationship among SbDof members with rice and Arabidopsis Dof proteins. An unrooted NJ tree is shown for 94 Dof proteins. The resulting groups are indicated as *A–F*. The *scale bar* corresponds to 0.05 estimated amino acid substitutions per site

The presence of many light responsive elements in the promoter region of *SbDof* genes strongly suggests the involvement of these *Dof* genes in regulation of photoperiodic control of flowering. The frequency of light responsive elements in *Dof* gene promoter region ranges from 1 to 6. The maximum number of light responsive elements i.e. 6 was found in the promoter region of *SbDof5* gene. The presence of *cis*-regulatory elements like Skn-1_motif, GCN4_motif, O2-site and RY element confers endosperm specific gene expression [34, 35]. The *SbDof19* and *SbDof24* contains GCN4_motif and Skn-1_motif and form cluster with *OsDof23* (*RPBF*) associated with the regulation of seed storage proteins [38]. Similarly *SbDof3*, *SbDof5*, *SbDof9*, *SbDof12*, *SbDof17*, *SbDof19*, *SbDof17*,

	AtDOF12	6 (5-24)	4 (30-59)	DOF (60-109)	38 (149-170)								
	AtDOF5	6 (5-24)	4 (36-63)	DOF (64-113)	38 (154-175)								
	OsDOF7	6 (11-30)	41 (42-68)	4 (69-96)	DOF (97-146)	12 (157-176)	10 (245-264)	18 (279-306)	7 (317-336)	2 (342-379)	3 (413-437)		
	SbDOF2	6 (10-29)	41(49-75)	4 (84-111)	DOF (112-161)	36 (169-202)	18 (302-329)	7 (357-376)	2 (379-416)	3 (451-475)			
	SbDOF23	6 (7-26)	41 (42-68)	4 (72-99)	DOF (100-149)	32 (155-174)	13 (179-206)	47 (233-270)	18 (275-302)	7 (313-332)	2 (336-373)	3 (400-424)	
	OsDOF6	6 (13-32)	41 (49-75)	4 (80-107)	DOF (108-157)	32 (163-182)	13 (188-215)	47 (245-282)	18 (290-317)	7 (328-347)	2 (351-388)	3 (416-440)	
	SbDOF14	6 (18-37)	4 (118-145)	DOF (146-195)	13 (233-260)	37 (268-287)	18 (375-402)	7 (412-431)	2 (435-472)	3 (488-512)			
	OsDOF4	6 (8-27)	4 (88-115)	DOF (116-165)	13 (211-238)	12 (249-268)	19 (274-312)	18 (352-379)	7 (389-408)	2 (412-449)	3(466-490)		
Group A	SbDOF15	6 (5-24)	26 (77-122)	4 (159-186)	DOF (187-236)	39 (242-268)	13 (283-310)	19 (342-380)	18 (393 -420)	7 (429-448)	2 (452 -485)	3 (506-530)	
	OsDOF3	6 (4-23)	26 (60-105)	4 (140-169)	DOF (170-219)	39 (225-251)	13 (267-294)	37 (302-321)	19 (328-366)	18 (410 -437)	7 (446-465)	2 (470 -507)	3 (527-551)
	SbDOF8	29 (11-35)	4 (45-72)	DOF (73-122)	13 (157-184)	19 (226-264)	18 (309-336)	7 (346-365)	2 (369-406)	3 (423-447)			
	OsDOF1	4 (82-109)	DOF (110-159)	13 (196-223)	19 (263-301)	18 (338-365)	7 (375-394)	2 (398-435)	3 (452-476)				
	AtDOF30	6 (2-21)	4 (112-139)	DOF (140-189)	13 (222-249)	19 (267-305)	18 (307-334)	7 (336-355)	2 (373-410)	3 (433-457)			
	AtDOF33	19 (90-128)	7 (134-153)	2 (158-195)	3 (213-237)								
	AtDOF17	6 (6-25)	41 (51-77)	4 (84-111)	DOF (112-161)	42 (215-242)	19 (290-328)	7 (336-355)	2 (365-402)	3 (424-448)			
	AtDOF9	6 (5-24)	20 (39-77)	4 (106-133)	DOF (134-183)	19 (238-276)	7 (286-305)	3 (373-397)					
	AtDOF3	6 (35-54)	4 (109-136)	DOF (137-186)	19 (242-280)	7 (285-304)	3 (372-396)						
	AtDOF26	22 (1-20)	DOF (27-76)	20 (77-115)	34 (123-149)	14 (154-181)	29 (186-210)						
Group B	AtDOF24	22 (1-20)	DOF (27-76)	20 (77-115)	34 (123-149)	14 (154-181)	29 (191-215)						
	AtDOF25	22 (1-20)	DOF (26-75)	20 (77-115)	34 (123-149)	14 (152-179)							
	AtDOF23	22 (1-20)	DOF (23-72)	20 (74-112)	34 (116-142)	14 (144-171)							
	SbDOF22	5 (46-65)	DOF (121-170)	12 (200-219)	48 (234-266)	27 (304-326)							
	OsDOF18	DOF (63-112)	12 (114-133)										
	SbDOF7	12 (17-36)	DOF (47-96)	27 (223-245)									
	OsDOF30	DOF (44-93)	12 (95-114)	27 (201-223)									
	OsDOF14	42 (1-28)	DOF (32-81)	33 (162-189)	12 (231-250)								
	SbDOF18	42 (1-28)	DOF (52-101)	12 (102-121)	33 (201-228)								
	OsDOF12	42 (1-28)	DOF (47-96)	12 (97-116)	33 (189-216)								
	SbDOF12	6 (4-23)	10 (103-122)	DOF (140-189)	18 (202-229)								
	OsDOF2	6 (10-29)	21 (54-73)	DOF (111-160)									
Group C	SbDOF1	DOF (46-95)	12 (104-123)	36 (183-216)									
	OsDOF27	DOF (53-102)	12 (107-126)	44 (241-260)									
	SbDOF9	12 (39-58)	DOF (103-152)	36 (270-303)									
	OsDOF19	DOF (35-84)	12 (90-109)	36 (111-144)	25 (218-237)								
	SbDOF13	6 (9-28)	DOF (87-136)	12 (142-161)									
	OsDOF20	6 (14-33)	DOF (86-135)	12 (141-160)									
	SbDOF10	DOF (74-123)	40 (240-284)										
	OsDOF28	12 (45-64)	DOF (81-130)	40 (234-278)	12 (302-321)								
	SbDOF27	DOF (31-80)	12 (88-107)	33 (185-212)									
	OsDOF15	DOF (32-81)	12 (92-111)	33 (166-193)									
	OsDOF23	DOF (47-96)	12 (97-116)	36 (260-293)									
	SbDOF24	9 (21-40)	DOF(41-90)	12 (91-110)	24 (123-171)	28 (197-244)	48 (250-282)	10 (298-317)					
	SbDOF19	9 (23-42)	DOF (43-92)	12 (93-112)	24 (126-174)	28 (200-247)	48 (253-285)						
	AtDOF6	DOF (31-80)	12 (83-102)										
	AtDOF36	DOF (34-83)	12 (85-104)										
Group D	AtDOF18	DOF (32-81)	12 (83-102)										
	SbDOF21	10 (37-56)	DOF (57-106)	12 (112-131)	18 (167-194)								
	OsDOF25	12 (4-23)	DOF (51-100)										
	OsDOF26	DOF (80-129)	12 (130-149)										
	SbDOF6	DOF (48-97)	12 (111-130)	49 (327-354)									
	OsDOF10	12 (22-41)	DOF (51-100)	49 (332-359)									
	AtDOF4	10 (4-23)	DOF (29-78)	12 (79-98)	37 (153-172)	49 (284-311)							
	AtDOF15	10 (5-24)	DOF (31-80)	12 (83-102)	45 (116-135)								
	AtDOF7	DOF (35-84)	12 (86-105)	45 (119-138)	21 (175-194)								
	SbDOF5	DOF (48-97)	12 (98-117)	8 (227-252)									
	OsDOF13	10 (20-39)	DOF (50-99)	12 (100-119)	8 (224-249)								
	AtDOF35	DOF (43-92)	8 (207-232)										
	AtDOF28	DOF (43-92)	8 (152-177)										
	OsDOF22	5(35-54)	DOF (55-104)	12 (105-124)	43 (138-176)	21 (243-262)	44 (275-294)	37 (313-332)					
	SbDOF4	5(30-49)	DOF (50-99)	12 (100-119)	43 (153-191)	10 (203-222)	21 (259-278)	44 (286-305)	37 (339-358)				
	OsDOF29	5(41-60)	DOF (61-110)	12 (111-130)	43 (162-200)	10 (210-229)	21 (263-282)	44 (290-309)	37 (336-355)				
Group E	SbDOF28	5 (73-92)	DOF (94-143)	12 (152-171)	10 (220-239)	21 (260-279)	37 (304-323)						
	OsDOF21	31 (1-20)	5 (75-94)	DOF (96-145)	12 (182-201)	10 (234-253)	21 (283-302)	37 (329-348)					
	SbDOF26	5 (28-47)	DOF (48-97)	12 (99-118)	12 (131-150)	10 (210-229)							
	OsDOF24	36(33-66)	5 (84-103)	DOF (104-153)	12 (155-174)	10 (254-273)							
	SbDOF3	DOF (19-68)	12 (69-88)	10 (172-191)	21 (231-250)	36 (271-304)							
	OsDOF5	31 (1-20)	5 (73-92)	DOF (93-142)	12 (143-162)	10 (258-277)	21 (319-338)	36 (345-378)					
	AtDOF20	31 (1-20)	5 (58-77)	DOF (78-127)	12 (128-147)	10 (214-233)	21 (249-268)						
	AtDOF29	31 (1-20)	10 (36-55)	5 (77-96)	DOF (97-146)	12(147-166)							
	AtDOF13	31 (1-20)	10 (32-51)	5 (71-90)	DOF (91-140)	29 (179-203)	21 (287-306)						
	AtDOF11	31 (1-20)	5 (75-94)	DOF (96-145)	21 (279-298)								
	AtDOF1	5 (3-22)	DOF (23-72)	12 (83-102)	21 (218-237)								
	AtDOF32	9 (27-46)	DOF (53-102)	12 (103-122)	49 (174-201)	6 (227-246)							
	AtDOF2	DOF (40-89)	17 (123-142)	32 (208-227)									
	AtDOF19	DOF (27-76)	12 (77-96)	17 (102-121)									
	SbDOF11	DOF (36-85)	12 (86-105)	30 (114-133)	17 (150-169)	35 (235-283)	15 (285-331)						
	OsDOF9	DOF (27-76)	12 (77-96)	30 (103-122)	17 (137-156)	35 (220-268)	15 (270-316)						
	SbDOF25	DOF (69-118)	12 (119-138)	30 (151-170)	17 (191-210)	15 (314-360)							
	OsDOF8	DOF (40-89)	30 (119-138)	17 (161-180)	12 (203-222)	15 (275-321)							
	AtDOF34	12 (43-62)	DOF (75-124)	36 (235-268)									
	SbDOF17	9 (62-81)	DOF (82-131)										
	OsDOF11	9 (45-64)	DOF (65-114)	12 (115-134)									
Group F	AtDOF10	21 (14-33)	DOF (50-99)	17 (139-158)	45 (234-253)	46 (258-277)							
	AtDOF31	DOF (57-106)	12 (107-126)	17 (142-161)	45 (199-218)	46 (232-251)							
	AtDOF16	DOF (42-91)	45 (190-209)	46 (219-238)									
	AtDOF22	DOF (70-119)	36 (254-287)										
	AtDOF21	32 (5-24)	DOF (76-125)	50 (161-180)	16 (252-271)								
	AtDOF14	DOF (70-119)	12 (120-139)	50 (162-181)	37 (232-251)	16 (291-310)	36 (318-351)						
	SbDOF16	9 (49-68)	DOF (69-118)	12 (119-138)	11(149-171)	25 (174-193)	23 (207-229)	33 (236-263)	16 (281-300)	36 (319-352)			
	OsDOF17	32 (2-21)	9 (62-81)	DOF (82-131)	12 (132-151)	11 (164-186)	25 (189-208)	23 (223-245)	33 (253-280)	16 (307-326)	36 (336-369)		
	SbDOF20	9 (43-62)	DOF (63-112)	12 (113-132)	11 (145-167)	25 (168-187)	23 (203-225)	33 (231-258)	16 (280-299)	36 (319-352)			
	OsDOF16	32 (2-21)	9 (60-79)	DOF (80-129)	12 (130-149)	11 (170-192)	25 (194-213)	23 (223-245)	33 (251-278)	16 (302-321)	36 (340-373)		
	AtDOF27	32 (1-20)	9 (35-54)	DOF (55-104)	7 (115-134)	11 (149-171)	44 (220-239)	16 (283-302)	36(306-339)				
	AtDOF8	32 (1-20)	9 (31-50)	DOF (51-100)	12 (101-120)	16 (276-295)	36 (308-341)						

◀ **Fig. 5** Schematic distribution of conserved motifs identified by means of MEME software among defined gene clusters. Position of each identified motif in all Dof proteins represented in *parenthesis*. Motif 1 highlighted with *black* represents the conserved Dof domain. Multilevel consensus sequences for the MEME defined motifs are listed in Table 4

SbDof24, *SbDof25* and *SbDof26* also revealed endosperm specific gene expression. The majority of the *SbDof* genes when subjected to *cis*-regulatory element analysis indicated their putative role in abiotic and biotic stress owing to the presence of stress responsive elements except *SbDof3*, *SbDof13*, *SbDof15*, *SbDof17* and *SbDof18*. The involvement of *SbDof* gene in regulation of meristem specific and hormone specific expression has also been observed. The list of identified *cis*-regulatory elements among the 26 *Dof* genes is shown in Table 5 and the putative functions of these elements are represented in Table 6. The diverse *cis*-regulatory elements present in *Dof* gene family might contribute to its multifarious roles.

Conclusions

In this study, we conducted a comprehensive computational analysis and identified a multitude *Dof* gene family in sorghum. A complete overview of this gene family in sorghum is presented, including the multiple sequence alignment, gene structures, phylogeny, chromosomal locations and their *cis*-regulatory elements analysis. In silico analysis revealed existence of 28 copies of C2C2-like *Dof* genes in *S. bicolor* genome. Multiple sequence alignment of these SbDof proteins showed well conserved four cysteine residues, putatively responsible for their zinc finger structures. Phylogenetic analysis further resulted into four subgroups constituting six clusters. The analysis of intron/exon gene structures revealed that most introns have conserved positions and phases, providing the evidence for the intron-early theory, and that multiple independent intron loss events likely have occurred during the evolution of flowering plants. The majority of predicted *SbDof* genes were intronless as observed in case of rice and Arabidopsis. The hypothesis that genome-wide and tandem duplication contributed to the expansion of the *Dof* gene family across the plant kingdom seems to be applicable for sorghum too. The comparative phylogenetic analysis of SbDof proteins with Arabidopsis and rice Dof proteins revealed six major groups. The SbDof members representing major group-A shows similarity with Arabidopsis CDF proteins involved in the regulation of photoperiodic control of flowering. The *SbDof19* and *SbDof24* shows similarity with rice PBF Dof

Table 4 Multilevel consensus sequences for the MEME defined motifs observed among different Dof proteins from sorghum, rice and Arabidopsis

Motif	No. of amino acids	E value	Multilevel consensus sequence
1	50	4.8e−4482	CPRCDSTNTKFCYYNNYNLSQPRHFCKTCRRYWTKGGALRNVPVGGGCRK
2	38	4.3e−253	EDEKREKKVWVPKTLRIDDPDEAAKSSIWTTLGIKPDD
3	25	1.5e−102	IIETLPVLQANPAALSRSQTFQETT
4	28	1.1e−095	DEKSEEEKTEGDESEQEKKLKKPDKILP
5	20	1.1e−092	MSMAERARLAKIPLPEPGLK
6	20	1.1e−085	DDPGIKLFGKTIPVPEDIEC
7	20	2.0e−038	ISIWCSDSNSFTLGKHSRDE
8	26	1.2e−033	SSIESLSFINQDLHWKLQQQRLATMF
9	20	1.9e−033	QQAAGGTERRARPQKEKALN
10	20	5.2e−032	AIGLEQWRVQQQQQQQPQQQ
11	23	1.2e−031	KNPKLMHEGAQDLNLAFPHHGGI
12	20	4.2e−031	NKRSKSSAAAAAAAAASAAA
13	28	5.8e−028	NETVLKFGPDVPLCESMASVLNIKEQNI
14	28	7.3e−027	HRLDFHDESFEQDYYDVGSDDLIVNQQI
15	47	4.2e−026	IVPPLTWWEDTKYDPFDSFPDDAMSLHDIMIGDEDHWWSVDCCQLEE
16	20	1.6e−025	DDEAAGRLLFPFEDLKPPVS
17	20	2.6e−024	DGSTIDLALLYAKFLNHHPD
18	28	8.9e−039	YIPIPPAVWWYWPCWPPGAWNAPWIRQN
19	39	2.9e−024	VDKEIWPYHGNCVMHPIPYYPGPPYMYPWNPAWNIVPVM
20	39	1.4e−023	KRPKIDQPSVAQMVSVEIQPGNHQPFKNVQENIDFVGSF
21	20	6.1e−023	KVSIVSGLITQLASVKMEDH

Table 4 continued

Motif	No. of amino acids	E value	Multilevel consensus sequence
22	20	6.6e−018	MDNLNVFANEDNQVNDVKPP
23	23	1.2e−017	VGALSAMELLRSTGCYMPLQQPM
24	49	2.7e−015	KMSINTQPMMVPNMMMPTPTMTGLFPNVLPTLMSTGEGGEFNFTMDNQH
25	20	1.8e−013	QPPEFPAFPSLESSSVCNPG
26	46	8.1e−011	KDDEIKVDVPQEEEDNEMKVDAPQEEKDDEMKIDVQEEKKDEEMEV
27	23	2.8e−010	FEWPSGCDLGTYWPTAVFADTDP
28	48	1.0e−009	TMPSFLEMLRKGLLHGSSSYDTGLVMSDGNNEMDMSFPLPAYGAMHGH
29	25	1.5e−009	YHMNQVDQFKWNQSFNNTMNMNYNN
30	20	1.3e−008	HRFPGPVRPDMILEGMVGNP
31	20	1.7e−015	MVFSSIPIYLDPPNWWQQQH
32	20	1.7e−008	MWTHWWQQLIVVKPMEEIIT
33	28	2.2e−008	YADQAFALGEFFAPPPPPPHPVLTFALD
34	27	5.1e−008	VTAVGNHFGYLSEIHGIMVTNPIPTFT
35	49	3.2e−006	TTATTTMLCTDVSVQAAFGELNFAMDQSCFDSLGLPTDDVVGGLLSSWC
36	34	1.0e−005	HHHHHDHHKEQDGGGGVIGGHETPGFWNGMIIGN
37	20	2.9e−008	SGWPWEDLSGFNSSSSGNVL
38	22	1.8e−005	HGGFRHDFPMKRLRCYTDGQSC
39	27	6.7e−005	LHYRQLLMAPDCMMGSRVEISKSMNPE
40	45	1.4e−004	HMDSQLGMGPLGQHDVLSSLGLKLPPPASSPPAASYYSDQLHAVV
41	27	1.7e−004	TDPKDQENTVQDSTDPPPQPEVVDTED
42	28	9.9e−006	MQEFQKIPGLAGRLFGGAAAADIVRAQE
43	39	2.9e−003	QLPFLASLHHPLGGGDHYSTGASRLGFPGLSSLDPVDYQ
44	20	7.0e−005	NSSPREFLGLPGNLQFWGGG
45	20	2.0e−005	NEQKQEMDPTRVLWGFPWQM
46	20	1.9e−003	HGHVDQIDSGREIWTNMVYI
47	38	2.5e−003	NQKEKVTAEKSPKIVQHPCMNGVAMWPFGCAPPPACYT
48	33	8.9e−004	TTDDARQLVGTQQGMNTDGGFVGSTRVQEEEEE
49	28	4.5e−003	YWGNGGIGGAAAAWPDLANCGSSIATLF
50	20	1.1e−002	DLNLLSFPVMQDQHHHHIEM

Table 5 Identified *cis*-regulatory elements of *SbDof* genes of sorghum, the position is shown in parenthesis

Annotated gene	Light responsive elements	Elements for endosperm specific gene expression	Hormone responsive elements	Elements for meristem specific expression	Stress responsive elements
SbDof2	I-box (−382)	Nil	Nil	Nil	ARE (−194)
	Sp1 (−248)				TCA-element (−11)
SbDof3	TCCC-motif (−428)	O2-site (−210)	ABRE (−388)	Nil	Nil
SbDof4	Gap-box (−341)	Nil	Nil	Nil	MBS (−366)
	I-box (−447)				
	MNF1 (−204)				
	Box II (−123)				
SbDof5	ATC-motif (−449)	RY-element (−409)	Nil	CCGTCC-box (−238)	CGTCA-motif (−181)
	Box 4 (−417)				TGACG-motif (−427)
	G-Box (−178)				
	MNF1 (−217)				
	Pc-CMA2c (−381)				
	Sp1(−341)				

Table 5 continued

Annotated gene	Light responsive elements	Elements for endosperm specific gene expression	Hormone responsive elements	Elements for meristem specific expression	Stress responsive elements
SbDof6	G-box (−56) Sp1 (−200) TCCC-motif (−193)	Nil	Nil	OCT (−134)	GC-motif (−272)
SbDof7	MNF1 (−148) Sp1 (−273)	Nil	Nil	CAT-box (−237)	MBS (−332) TGACG-motif (−232)
SbDof8	ACE (−61) TCT-motif (−89)	Nil	Nil	Nil	Box-W1 (−125) MBS (−451)
SbDof9	G-box (−473) GATA-motif (−481)	RY-element (−425)	ABRE (−473)	CAT-box (−46)	GC-motif (−121) ARE (−309) MBS (−361) CGTCA-motif (−68)
SbDof10	G-box (−245) I-box (−84) Sp1 (−131)	Nil	Nil	Nil	ARE (−398) GC-motif (−463)
SbDof11	G-Box (−446) Sp1 (−66) TCCC-motif (−174)	Nil	ABRE (−19)	Nil	GC-motif (−439) CGTCA-motif (−294)
SbDof12	G-box (−8) Sp1 (−337)	GCN4_motif (−129)	P-box (−322) TGA-element (−164)	CCGTCC-box (−220)	TGACG-motif (−31)
SbDof13	Sp1 (−45)	Nil	AuxRR-core (−326)	Nil	Nil
SbDof14	Sp1 (−304) TCCC-motif (−132) rbcS-CMA7a (−259)	Nil	Nil	CCGTCC-box (−94)	MBS (−385) GC-motif (−233) TCA-element (−269)
SbDof15	ATCC-motif (−213) GT1-motif (−386)	Nil	Nil	Nil	Nil
SbDof17	Sp1 (−490)	RY-element (−141) Skn-1_motif (−359)	Nil	Nil	Nil
SbDof18	G-Box (−42) Sp1 (−120)	Nil	Nil	Nil	Nil
SbDof19	ATCT-motif (−387) G-Box (−226)	GCN4_motif (−422) Skn1_motif (−419)	TGA-element (−283)	Nil	ARE (−431)
SbDof20	Pc-CMA2c (−155) Sp1 (−194)	O2-site (−61)	Nil	Nil	TCA-element (−346)
SbDof21	CATT-motif (−235) Sp1 (−392)	Nil	TGA-element (−358)	CAT-box (−242) CCGTCC-box (−35)	CGTCA-motif (−462) TGACG-motif (−286)
SbDof22	Box I (−165) G-box (−69) Sp1 (−77)	Nil	ABRE (−30) TGA-box (−364)	CCGTCC-box (−176)	CCAAT-box (−5) C-repeat/DRE (−324) CGTCA-motif (−67) TGACG-motif (−364)
SbDof23	Box II (−310) G-Box (−309) Sp1 (−293) Box II (−311)	Nil	ABRE (−308)	CCGTCC-box (−250)	GC-motif (−290) CGTCA-motif (−462)
SbDof24	Box 4 (−64) TCT-motif (−52)	GCN4_motif (−305) Skn-1_motif (−302)	Nil	Nil	ARE (−164)

Table 5 continued

Annotated gene	Light responsive elements	Elements for endosperm specific gene expression	Hormone responsive elements	Elements for meristem specific expression	Stress responsive elements
SbDof25	G-Box (−303) G-box (−199) L-box (−306)	Skn-1_motif (−342)	ABRE (−363) P-box (−225)	CCGTCC-box (−24)	CGTCA-motif (−343)
SbDof26	Sp1 (−205)	RY-element (−177)	Nil	Nil	MBS (−407)
SbDof27	G-box (−341) Sp1 (−115) TGG-motif (−395)	Nil	Nil	Nil	MBS (−77)
SbDof28	Box 4 (−32)	Nil	Nil	Nil	MBS (−472) ARE (−256)

Table 6 Functions of identified *cis*-regulatory elements

S. No.	*Cis*-regulatory elements	Functions/description
1	ARE	*Cis*-acting regulatory element essential for the anaerobic induction
2	Box-W1	Fungal elicitor responsive element
3	CCGTCC-box	*Cis*-acting regulatory element related to meristem specific activation
4	GC-motif	Enhancer-like element involved in anoxic specific inducibility
5	GCN4_motif	*Cis* -regulatory element involved in endosperm expression
6	MBS	MYB binding site
7	MBSII	MYB binding site involved in flavonoid biosynthetic genes regulation
8	Skn-1_motif	*Cis*-acting regulatory element required for endosperm expression
9	ABRE	*Cis*-acting element involved in the abscisic acid responsiveness
10	CGTCA-motif	*Cis*-acting regulatory element involved in the MeJA-responsiveness
11	TGACG-motif	*Cis*-acting regulatory element involved in the MeJA-responsiveness
12	GARE-motif	Gibberellin-responsive element
13	Sp1	Light responsive element
14	Box 4	Part of a conserved DNA module involved in light responsiveness
15	I-box	Light responsive element
16	O2-site	*Cis*-acting regulatory element involved in zein metabolism regulation
17	MNF1	Light responsive element
18	ATC-motif	Part of a conserved DNA module involved in light responsiveness
19	G-Box	*Cis*-acting regulatory element involved in light responsiveness
20	Pc-CMA2c	Part of a light responsive element
21	RY-element	*Cis*-acting regulatory element involved in seed-specific regulation
22	OCT	*Cis*-acting regulatory element related to meristem specific activation
23	CAT-box	*Cis*-acting regulatory element related to meristem expression
24	ACE	*Cis*-acting element involved in light responsiveness
25	TCT-motif	Part of a light responsive element
26	GATA-motif	Part of a light responsive element
27	TCCC-motif	Part of a light responsive element
28	P-box	Gibberellin-responsive element
29	TGA-element	Auxin-responsive element
30	AuxRR-core	*Cis*-acting regulatory element involved in auxin responsiveness
31	TCA-element	*Cis*-acting element involved in salicylic acid responsiveness
32	rbcS-CMA7a	Part of a light responsive element
33	ATCC-motif	Part of a conserved DNA module involved in light responsiveness
34	GT1-motif	Light responsive element
35	ATCT-motif	Part of a conserved DNA module involved in light responsiveness

Table 6 continued

S. No.	*Cis*-regulatory elements	Functions/description
36	Pc-CMA2c	Part of a light responsive element
37	Box I	Light responsive element
38	C-repeat/DRE	Regulatory element involved in cold- and dehydration-responsiveness
39	CCAAT-box	MYBHv1 binding site
40	Box II	Part of a light responsive element
41	L-box	Part of a light responsive element
42	TGG-motif	Part of a light responsive element

in group-D which was further confirmed by the presence of seed storage protein specific regulatory elements like GCN4_motif and Skn-1_motif.

The comparative phylogenetic analysis with respect to the *Dof* gene family clearly indicates the proximity of sorghum with rice than Arabidopsis owing to the fact that sorghum and rice are members of monocot while Arabidopsis is a dicot. Further the presence of some similar groups and sub groups in comparative phylogeny of sorghum, rice and Arabidopsis revealed the conserved nature of some of the *Dof* genes during angiosperm evolution. The uniform presence of motif 1 representing the conserved Dof domain further supports the identity of the predicted genes as *Dof* gene family in sorghum. The *cis*-regulatory element analysis of the predicted *Dof* genes revealed the major putative functions as regulation of genes associated with seed storage proteins, abiotic and biotic stress, photoperiod, growth hormone and meristem. The in silico analysis of the predicted *SbDof* gene family needs to be supported by wet lab experiments by performing cloning, sequencing and expression profiling of the respective genes.

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